Jasmine Kaur

Punjab, India | +91 7814213978 | jasminekaur3k@gmail.com | linkedin.com/in/jasmine-kaur-grewal | github.com/jasminegrewal05 | Portfolio

Computer Science undergraduate building data pipelines, risk models, and dashboards with SQL, Python, and Azure.

Education

Thapar Institute of Engineering and Technology — Patiala, India
B.E. Computer Science and Business Systems — *Aug 2024 – May 2028 (Expected)*
- CGPA: 8.71/10 | Branch Rank: 15
- Relevant Coursework: DBMS, Cloud Computing, Data Structures, Statistics, Statistical Modeling

Skills

Languages: Python, SQL, C++, C
Databases & Cloud: PostgreSQL, Redis, Microsoft Azure (Azure Storage)
Data Analysis & Visualization: Pandas, NumPy, EDA, Feature Engineering, Matplotlib, Seaborn, Streamlit, Excel
Machine Learning: Scikit-learn, XGBoost, Random Forest, SVM, Logistic Regression, SHAP
Tools: Git, GitHub, Jupyter, VS Code

Experience

Undergraduate Research Intern — Feb 2026 – Present
Thapar Institute of Engineering and Technology — *Patiala, India*
- Developed a Type 2 Diabetes screening pipeline on 16 clinical biomarkers, comparing 6 ML models (ROC-AUC 0.84).
- Tuned thresholds for recall, cutting missed cases by 66; presented at AITA 2026, an international AI conference.
- Managed research datasets on Azure Storage for model training and evaluation.

Astronomy & Astrophysics Intern — Jun 2025 – Jul 2025
India Space Academy — *Remote*
- Analyzed SDSS spectroscopic data to estimate a galaxy cluster's dynamical mass and its dark matter share.

Projects

CITY-AI: Multi-Camera ANPR & Traffic Analytics | *PostgreSQL, PyTorch, React* | Demo — Sep 2026
- Designed a PostgreSQL schema storing real-time detection events and vehicle tracks from live camera feeds, fed by a Redis event pipeline and queried with SQL for a React GIS dashboard.
- Trained a CRNN+CTC license-plate recognizer from scratch on 1,059 annotated crops and 75,000 synthetic plates, reaching 93.6% exact-match and 98.8% character accuracy on a leak-free group-split test set.
- Raised accuracy on degraded frames from 69.1% to 90.4% using confidence-weighted multi-frame voting over ByteTrack tracks, with a four-tier read system that flags obscured plates.

Credit Risk & Portfolio Optimization Engine | *XGBoost, SHAP, Streamlit* | GitHub | Live Demo — May 2026
- Modeled probability of default for 30,000 accounts with XGBoost (ROC-AUC 0.82), using reject inference and SHAP.
- Engineered behavioral and delinquency features; built 3 risk bands driving automated credit limit recommendations.

Cloud-Based Resume Screening Platform | *Microsoft Azure* — In Progress
- Building a recruiter platform on Azure that ingests resumes and screens candidates against job requirements.
- Parsing resumes into structured candidate records so applicants can be compared consistently.

Achievements & Certifications

McKinsey Forward Learning Program, McKinsey & Company, 2026
Semi-Finalist, Flipkart GRiD 2026, a national engineering and business case challenge
Finalist, HackVega 2026
District-Level Athlete in Field Athletics and Basketball

Leadership & Volunteering

Design Coordinator, *Microsoft Learn Student Chapter (MLSC), TIET* — Jun 2025 – Present
- Led visual design and marketing for developer events serving 100+ members; screened 30+ recruitment candidates.

Marketing Core Member, *Saturnalia 2025, Thapar University Annual Fest* — Sep 2025 – Nov 2025
- Pursued sponsors through cold calls, emails, and in-person visits, contributing to partner acquisition.

Volunteer Educator, *Pratigya Abhiyan, Patiala* — Oct 2024 – Sep 2025
- Taught weekly Math, English, and Science to 30+ underprivileged students for 11 months; ran STEM workshops.